

INTRODUCTION

Most of the new drugs developed by the pharmaceutical industry are poorly water-soluble in gastro-intestinal fluids which impair their oral bioavailability. Innovative lipid-based oral delivery systems are known to improve drug oral bioavailability. However, there is a lack of understanding of the mechanisms underlying the bioavailability enhancement effects of these formulations. The digestion of lipid-based formulation by lipases will have a major influence on the fate of the drug in the gut [1]. Labrasol® is a caprylocaproyl macrogolglyceride which forms oil-in-water microemulsion in water or in gastro-intestinal fluids [2]. Labrasol® is composed of mono, di and triglycerides, and PEG-8 mono and diesters, potential substrates of digestive lipases.

BACKGROUND

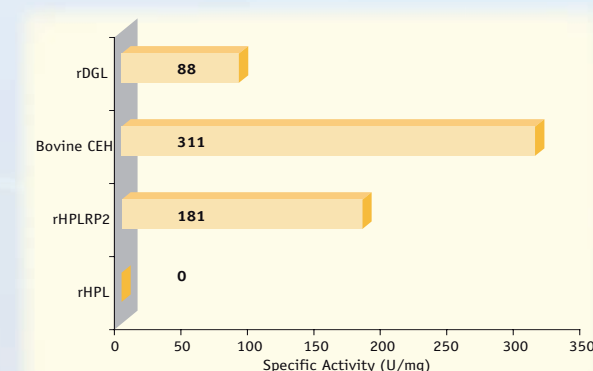


Figure 1: Specific activities of digestive lipases on Labrasol®

We have previously shown the *in-vitro* lipolysis of Labrasol® (Figure 1) by several purified digestive lipases [3].

Human pancreatic lipase (HPL), the main lipase involved in the digestion of dietary triglycerides [4, 5], does not hydrolyse Labrasol® contrary to other pancreatic lipases such as human pancreatic lipase-related protein 2 (HPLRP2) and bovine carboxyl ester hydrolase (CEH).

Dog gastric lipase (DGL) is also able to hydrolyse Labrasol® which suggests that *in-vivo*, digestion of Labrasol® starts within the stomach.

PURPOSES

- Identify the composition changes during *in-vitro* gastric and pancreatic lipolysis of Labrasol®.
- Evaluate the relative contribution of gastric and duodenal digestion steps.
- Determine if porcine pancreatic extracts are a good model for human pancreatic juice.

METHODS

Experimental conditions were based on *in-vivo* data at 50% gastric emptying of the meal, both in the stomach and in the duodenum.

In-vitro lipolysis were performed in a temperature-controlled reaction vessel at 37°C for 30 min at pH 5.5 with DGL (gastric digestion step) and then for 60 min with either porcine pancreatic extract (PPE) or human pancreatic juice (HPJ) (duodenal digestion step) at pH 6.25.

The pH was kept constant via the automated titration of free fatty acids released with 0.1 M NaOH solution (Figure 2). The biorelevant medium used was described in [3]. Samples were collected and analysed by HPLC and GC to assay glycerides and PEG esters.

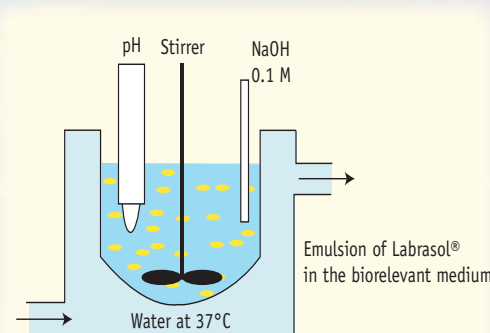


Figure 2: Schematic representation of pH-stat device

RESULTS

Contrary to monoglycerides, di and triglycerides are rapidly hydrolysed by DGL (Figures 3A and 3C). PEG diesters were quickly hydrolysed by DGL whereas there was no lipolysis of PEG monoesters. The lipolysis of PEG diesters corresponds to a slight increase of PEG monoesters and the apparition of free PEG (Figures 3B and 3D). PEG diesters are better substrates for DGL than PEG monoesters.

DGL showed a clear preference for insoluble or partially soluble substrates such as medium chain triglycerides, diglycerides and PEG diesters. It is necessary to consider the gastric digestion step because, *in-vivo*, Labrasol® is probably hydrolysed by gastric lipase.

All compounds of Labrasol® are totally hydrolysed 10 min and 15 min after addition of either PPE (Figures 3A and 3B) or HPJ (Figures 3C and 3D), respectively.

Five minutes after the addition of pancreatic lipases, PEG diesters were totally hydrolysed whereas the lipolysis of PEG monoesters was completed after 10 min for PPE and after 15 min for HPJ. At 60 min, the percentage of PEG monoesters raised up to 8% and 10% for porcine pancreatic extract and human pancreatic juice, respectively.

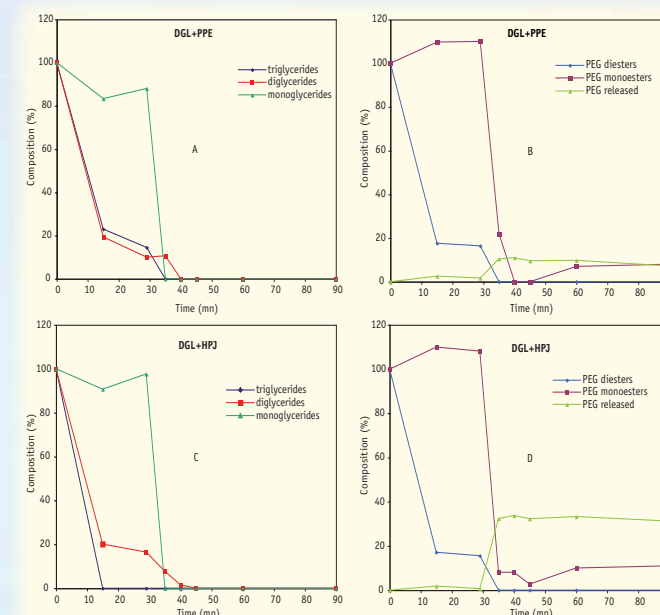


Figure 3: Composition changes during *in-vitro* lipolysis of Labrasol® with dog gastric lipase (DGL) for the gastric digestion step and, either porcine pancreatic extracts (PPE) (A and B) or human pancreatic juice (HPJ) (C and D) for the pancreatic digestion step.

CONCLUSIONS AND PROSPECTS

Labrasol® is hydrolysed in less than 45 min by both gastric and pancreatic lipases. The gastric digestion step is responsible for most of the lipolysis of diglycerides, triglycerides and PEG diesters. DGL showed a clear-cut preference for di and triglycerides and PEG diesters compared to monoglycerides and PEG monoesters. The lipolytic activity of DGL decreases over time because of an inhibition by lipolytic by-products. Concerning the duodenal digestion step, porcine pancreatic extract and human pancreatic juice present the same lipolytic profiles. Although porcine pancreatic extract can be used as a model of human pancreatic juice, it is necessary to consider the gastric digestion step which significantly modifies the initial composition of Labrasol®.

It would be interesting now to investigate the *in-vitro* lipolysis of a formulation containing Labrasol® and a poorly water-soluble drug to evaluate the impact of both gastric and duodenal digestion steps on the solubility of that model drug.

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